HDOC DAY-67

Write a recursive function to find the **GCD of two positive integers using Euclid's algorithm**.

Algorithm

1. Start.
2. Input two positive integers a and b.
3. Call gcd(a, b).
4. In the gcd() function:
    - If b == 0, return a. This is the **base case**.
    - Otherwise, recursively call gcd(b, a % b).
5. Continue until the second number becomes 0.
6. The remaining first number is the GCD.
7. Display the GCD.
8. Stop.

- ## C Program:

```
#include <stdio.h>

// Recursive function to find GCD
int gcd(int a, int b)
{
    if (b == 0)
        return a;           // Base case

    return gcd(b, a % b);     // Recursive call
}

int main()
{
    int a, b;

    printf("Enter two positive integers: ");
    scanf("%d %d", &a, &b);

```

```
    printf("GCD of %d and %d = %d\n", a, b, gcd(a, b));

    return 0;
}
```

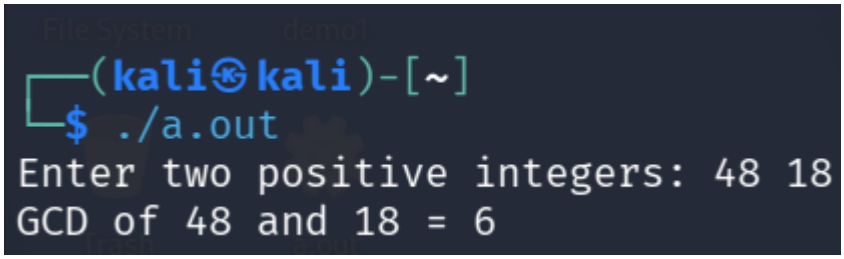
Test Case Table:

Test Case	Input a	Input b	Expected GCD	Result
1	48	18	6	Pass
2	20	15	5	Pass
3	100	25	25	Pass
4	17	13	1	Pass
5	7	7	7	Pass

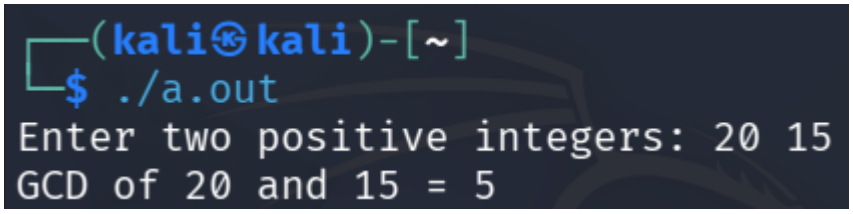
Compilation and Execution:

The following program has been compiled and executed successfully.

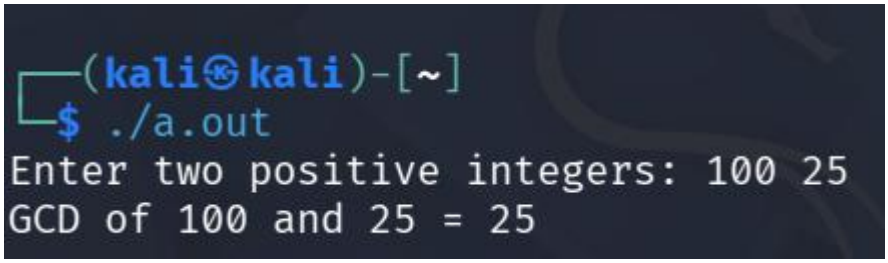
Test Case 1:



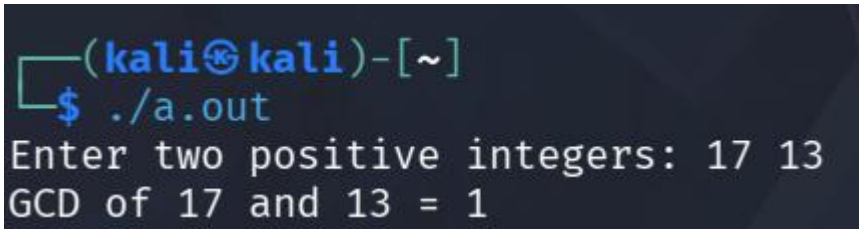
Test Case 2:



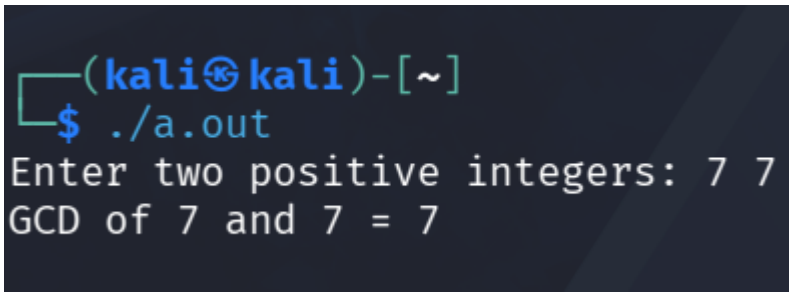
Test Case 3:



Test Case 4:


(kali㉿kali)-[~]
$ ./a.out
Enter two positive integers: 17 13
GCD of 17 and 13 = 1

Test Case 5:


(kali㉿kali)-[~]
$ ./a.out
Enter two positive integers: 7 7
GCD of 7 and 7 = 7